

# **Litmus Migration Guide**

**Product Version 26.10**

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# Contents

|                                              |    |
|----------------------------------------------|----|
| 1                                            | 4  |
| Commands / Rule Attributes Mapping           | 4  |
| 2                                            | 7  |
| Root Attribute / Global Variable Mapping     | 7  |
| 3                                            | 8  |
| Incremental Constraints (SDC) Support        | 8  |
| Example SDC                                  | 8  |
| Example Non-MMMC flow                        | 8  |
| Example MMMC flow                            | 9  |
| 4                                            | 12 |
| Save / Restore Session                       | 12 |
| 5                                            | 13 |
| Incremental Constraints with Restore Session | 13 |
| 6                                            | 14 |
| Lint Configuration Support                   | 14 |
| 7                                            | 15 |
| Reduced Pessimism in Litmus                  | 15 |
| CONST_CLK_GRP3                               | 15 |
| CONST_DGN_CMB1 and CONST_EXC_SMD7            | 16 |
| CONST_DGN_CMB3                               | 17 |
| CONST_EXC_MCP5                               | 18 |
| CONST_EXC_MCP7                               | 18 |
| 8                                            | 20 |
| Rule Behavior Changes                        | 20 |
| CONST_CG_CHK3                                | 22 |
| CONST_CLK_DEF1                               | 23 |
| CONST_CLK_DEF2                               | 23 |
| CONST_CLK_DEF40                              | 23 |
| CONST_CLK_GRP2                               | 24 |

|                                |    |
|--------------------------------|----|
| CONST_CLK_GRP4                 | 24 |
| CONST_CLK_UNC10                | 24 |
| CONST_IO_DRC1                  | 24 |
| CONST_IO_IDL1                  | 24 |
| CONST_IO_IDL13                 | 24 |
| CONST_IO_ITR1                  | 24 |
| 9                              | 25 |
| New Rules in Litmus            | 25 |
| 1D2S (1 Design, 2 SDC)         | 25 |
| 10                             | 26 |
| Deprecated rules in Litmus     | 26 |
| 11                             | 27 |
| Launching GUI                  | 27 |
| Steps to Launch GUI            | 27 |
| 12                             | 30 |
| Shell Default Behavior Changes | 30 |
| 13                             | 31 |
| SVA Generation                 | 31 |
| What is supported?             | 31 |
| What is not supported?         | 31 |
| Command Line Changes           | 31 |
| How to enable Mode1 & Mode2    | 32 |
| Changes to control commands    | 32 |

## Commands / Rule Attributes Mapping

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| Conformal Litmus Command                              | Litmus Equivalent                                                                                                                                                                                                                                                                                               | Usage Notes                                                                                                                                                       |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| read_libs                                             | read_libs                                                                                                                                                                                                                                                                                                       | Same as Conformal Litmus                                                                                                                                          |
| read_netlist                                          | read_netlist                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                   |
| read_sdc                                              | read_sdc                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                   |
| init_design                                           | init_design                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                   |
| add_notranslate_modules                               | Not Supported                                                                                                                                                                                                                                                                                                   | manually skip module definition for the purpose                                                                                                                   |
| report_library_data                                   | report_reference                                                                                                                                                                                                                                                                                                | -skip_unref                                                                                                                                                       |
| set_message -suppress / -unsuppress                   | set_rule_handling [ -suppress   -unsuppress ]                                                                                                                                                                                                                                                                   | If a rule is suppressed, its severity will be "Ignore".<br><br>When a suppressed rule is unsuppressed, its severity will be changed back to its default severity. |
| report_design_data                                    | report_netlist_statistics                                                                                                                                                                                                                                                                                       |                                                                                                                                                                   |
| report_black_box                                      | report_reference<br>get_db insts -if { .is_black_box == true<br>}                                                                                                                                                                                                                                               |                                                                                                                                                                   |
| write_lint_configuration<br>report_lint_configuration | write_lint_configuration<br>report_lint_configuration                                                                                                                                                                                                                                                           |                                                                                                                                                                   |
| write_db<br>read_db                                   | write_rule_db<br>read_rule_db                                                                                                                                                                                                                                                                                   |                                                                                                                                                                   |
| add_rule_attribute <> -rule <rule>                    | define_attribute <> -obj_type<br>rule_occurrence -data_type string -category<br><> default -compute_function <><br><br>User-defined attribute is supported on rule container object:<br><br>define_attribute <> -obj_type const_exc_flp1<br>-data_type string -category <> -default <> -<br>compute_function <> | const_exc_flp1 is rule container object for CONST_EXC_FLP1 rule.                                                                                                  |

**Litmus Migration Guide**  
Commands / Rule Attributes Mapping

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|                                                                                                                                                             |                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| define_attribute <> -obj_type rule_occurrence -data_type string -default ""                                                                                 | define_attribute <> -obj_type rule_occurrence -data_type string -category <> -default <><br>User-defined attribute is supported on rule container object.<br>Supported by : define_attribute <> -obj_type const_exc_flp1 -data_type string -category <> -default <> -compute_function <> | const_exc_flp1 is rule constrainer object for CONST_EXC_FLP1 rule                                                                                                                  |
| set_rule_property -rule <> -table_header <>                                                                                                                 | set_db <rule_instance> .ltms_table_header <>                                                                                                                                                                                                                                             | User-defined attribute on rule container object is automatically added into ltms_table_header attribute of rule instance object, also available in spreadsheet output.             |
| set_db <rule_instance> .cfm_table_header <>                                                                                                                 | set_db <rule_instance> .ltms_table_header <>                                                                                                                                                                                                                                             | User-defined attribute on rule container object is automatically added into ltms_table_header attribute of rule instance object, also available in spreadsheet output.             |
| get_db rule_instance.*                                                                                                                                      | get_db rule_instances<br>get_db rule_instances *                                                                                                                                                                                                                                         |                                                                                                                                                                                    |
| get_db <rule_instance> .cfm_rule_occurrences                                                                                                                | get_db <rule_instance> .ltms_rule_occurrences                                                                                                                                                                                                                                            |                                                                                                                                                                                    |
| get_db <rule_instance> -if { .cfm_is_run==1 }                                                                                                               | get_db <rule_instance> -if { .ltms_is_run == true }                                                                                                                                                                                                                                      |                                                                                                                                                                                    |
| get_db <rule_instance> -if { .cfm_status==Fail }                                                                                                            | get_db <rule_instance> -if { .ltms_status == Fail }                                                                                                                                                                                                                                      |                                                                                                                                                                                    |
| get_db <rule_instance> -if { .cfm_rule_filters==*<>* }                                                                                                      | get_db <rule_instance> -if { .ltms_rule_filters == *<>* }                                                                                                                                                                                                                                |                                                                                                                                                                                    |
| get_db <rule_occurrence> .cfm_is_filtered                                                                                                                   | get_db <rule_occurrence> .ltms_is_filtered                                                                                                                                                                                                                                               |                                                                                                                                                                                    |
| get_db <rule_occurrence> .cfm_filtered_by                                                                                                                   | get_db <rule_occurrence> .ltms_filtered_by                                                                                                                                                                                                                                               |                                                                                                                                                                                    |
| get_db <rule_occurrence> .cfm_constraint_objects<br>get_db <constraint_object> .cfm_constraint_statements<br><br>get_db <constraint_statement> .<attribute> | get_db <rule_occurrence> .rule_info.*<br>file_name<br>file_line<br>exc_from<br>exc_through<br>exc_to<br>exc_type                                                                                                                                                                         | Constraint Object is not available in Litmus, hence attributes .cfm_constraint_objects .cfm_constraint_statements are unavailable. Litmus keeps relevant info in .rule_info object |
| verify_design_modeling                                                                                                                                      | Not Supported                                                                                                                                                                                                                                                                            |                                                                                                                                                                                    |

**Litmus Migration Guide**  
Commands / Rule Attributes Mapping

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|                                                                                                                                                                                          |                                                                                                                                                                 |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| add_rule_filter<br>XXX_CONST_CG_CHK3_cg_check_mux_DtoD -rule<br>CONST_CG_CHK3 -filter<br>"clock_gating_enable_pin.inst.cfm_type == MUX &&<br>clock_gating_enable_pin.cfm_type == DATAIN" | add_rule_filter <> -rule CONST_CG_CHK3 -<br>filter "clock_gating_enable_pin.inst.is_mux<br>== true &&<br>clock_gating_enable_pin.is_mux_select_pin<br>== false" |  |
| add_rule_filter XXX_CONST_CLK_DEF6_filter_mux -<br>rule CONST_CLK_DEF6 -filter<br>"clock_point.inst.cfm_type == MUX"                                                                     | add_rule_filter <> -rule CONST_CLK_DEF6<br>-filter "clock_point.inst.is_mux == true"                                                                            |  |
| add_rule_filter<br>XXX_CONST_DGN_TIM1_filter_mux2_0 -rule<br>CONST_DGN_TIM1 -filter<br>"cfm_design_objects.base_name == mux2_0"                                                          | add_rule_filter <> -rule CONST_DGN_TIM1 -<br>filter "mux.base_name == <>"                                                                                       |  |
| add_rule_filter<br>XXX_CONST_DGN_TIM1_filter_mux_clk_select -rule<br>CONST_DGN_TIM1 -filter "is_tied == 1"                                                                               | add_rule_filter <> -rule CONST_DGN_TIM1 -<br>filter "is_tied == true"                                                                                           |  |
| add_rule_filter<br>XXX_CONST_LINT_IDL7_filter_seq_clk -rule<br>CONST_LINT_IDL7 -filter<br>{cfm_design_objects.inst.cfm_type == DFF &&<br>cfm_design_objects.cfm_type == CLOCK}           | add_rule_filter <> -rule CONST_LINT_IDL7 -<br>filter {pin.inst.is_flop == true && pin.is_clock<br>== true}                                                      |  |
| add_rule_filter<br>XXX_CONST_LINT_ODL8_filter_seq_d -rule<br>CONST_LINT_ODL8 -filter<br>{cfm_design_objects.inst.cfm_type == DFF &&<br>cfm_design_objects.cfm_type == DATA}              | add_rule_filter <> -rule CONST_LINT_ODL8<br>-filter {pin.inst.is_flop == true && pin.is_clock<br>== false}                                                      |  |
| add_rule_filter f1 -rule CONST_CLK_DEF23 -filter<br>"original_clock_command=~*create_generated_clock**"                                                                                  | add_rule_filter f1 -rule CONST_CLK_DEF23<br>-filter "original_clock.is_generated==true"                                                                         |  |

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## Root Attribute / Global Variable Mapping

---

In general, all Conformal Litmus settings prefixed *cfm* are available as the same name prefixed with *ltms* in Litmus. For example, `cfm_const_report_all_witnesses` is available as `ltms_const_report_all_witnesses`.

Example:

| Conformal Litmus                                                  | Litmus                                                             |
|-------------------------------------------------------------------|--------------------------------------------------------------------|
| <code>root/cfm_const_multicycle_path_multiplier_difference</code> | <code>root/ltms_const_multicycle_path_multiplier_difference</code> |
| <code>root/cfm_const_input_delay_ratio</code>                     | <code>root/ltms_const_input_delay_ratio</code>                     |
| <code>root/cfm_const_output_delay_ratio</code>                    | <code>root/ltms_const_output_delay_ratio</code>                    |
| <code>root/cfm_const_report_all_witnesses</code>                  | <code>root/ltms_const_report_all_witnesses</code>                  |
| <code>root/cfm_const Consolidate occurrences</code>               | <code>root/ltms_const Consolidate occurrences</code>               |



---

# Incremental Constraints (SDC) Support

---

Litmus introduces support for providing incremental constraints and subsequently performing rule checking in the same session. This was not possible with Conformal Litmus.

## Example SDC

```
create_clock -name CLK1 -period 5.000 -waveform [list 0.000 2.500] [get_ports clk1]
create_clock -name CLK2 -period 15.000 -waveform [list 0.000 2.500] [get_ports clk2]
set_sense -stop_propagation buf4/X -clocks CLK1

set_input_delay -clock CLK1 1.0 [ get_ports data1]
set_input_delay -clock CLK2 1.0 [ get_ports data2]
```

## Example Non-MMMC flow

### Main Litmus Script

```
set libs [list lib1.lib lib2.lib]
read_lib $libs
read_netlist top.v
init_design
read_sdc top.sdc
verify_timing_intent
```

### Output

```
@[DEV]litmus 12> report_rule_check CONST_CLK_LAT1 -verbose
CONST_CLK_LAT1: Undefined clock network latency for real and generated clocks
Severity: Warning      Fail Occurrence: 2
1: Fail: In line 1, file top.sdc (create_clock): Object: CLK1
2: Fail: In line 2, file top.sdc (create_clock): Object: CLK2
```

### Incremental SDC constraint

```
@[DEV]litmus > set_clock_latency 0.3 [get_clocks CLK1]
@[DEV]litmus > verify_timing_intent
...
@[DEV]litmus > report_rule_check CONST_CLK_LAT1 -verbose
CONST_CLK_LAT1: Undefined clock network latency for real and generated clocks
Severity: Warning      Fail Occurrence: 1
1: Fail: In line 2, file top.sdc (create_clock): Object: CLK2
```

## Example MMC flow

### MMMC script (mmmc.tcl)

```
create_library_set -name max_libset -timing [list lib1.lib lib2.lib]
create_library_set -name min_libset -timing [list minlib1.lib minlib2.lib]

create_timing_condition -name default_mapping_tc_2 -library_sets [list max_libset]
create_timing_condition -name default_mapping_tc_1 -library_sets [list min_libset]

create_rc_corner -name rc_corner_worst\
  -pre_route_res 1\
  -post_route_res 1\
  -pre_route_cap 1\
  -post_route_cap 1\
  -post_route_cross_cap 1\
  -pre_route_clock_res 0\
  -pre_route_clock_cap 0\
  -temperature 0

create_rc_corner -name rc_corner_best\
  -pre_route_res 1\
  -post_route_res 1\
  -pre_route_cap 1\
  -post_route_cap 1\
  -post_route_cross_cap 1\
  -pre_route_clock_res 0\
  -pre_route_clock_cap 0\
  -temperature 25

create_delay_corner -name min_delay_corner -timing_condition {default_mapping_tc_1} -
rc_corner rc_corner_worst

create_delay_corner -name ocv_delay_corner -early_timing_condition
{default_mapping_tc_1} -late_timing_condition {default_mapping_tc_2} -rc_corner
rc_corner_worst

create_delay_corner -name max_delay_corner -timing_condition {default_mapping_tc_2} -
rc_corner rc_corner_best

create_constraint_mode -name CM1 -sdc_files[list ./top.sdc]

create_analysis_view -name view1 -constraint_mode CM1 -delay_corner ocv_delay_corner

set_analysis_view -setup [list view1] -hold [list view1]
```

## Main Litmus Script

```
read_mmmc mmmc.tcl

read_netlist ./top.v

init_design

verify_timing_intent
```

## Output

```
@[DEV]litmus > report_rule_check CONST_CLK_LAT1 -verbose
CONST_CLK_LAT1: Undefined clock network latency for real and generated clocks
  Severity: Warning    Fail Occurrence: 2
  1: Fail: In line 1, file top.sdc (create_clock): Object: CLK1
  2: Fail: In line 2, file top.sdc (create_clock): Object: CLK2
```

## Incremental SDC Constraint

```
@[DEV]litmus > set_interactive_constraint_modes CM1
@[DEV]litmus > set_clock_latency 0.3 [get_clocks CLK1]
@[DEV]litmus > verify_timing_intent
...
@[DEV]litmus 16> report_rule_check CONST_CLK_LAT1 -verbose
CONST_CLK_LAT1: Undefined clock network latency for real and generated clocks
  Severity: Warning    Fail Occurrence: 1
  1: Fail: In line 2, file top.sdc (create_clock): Object: CLK2
```

---

## Save / Restore Session

---

The following commands should be used to save or restore a Litmus session:

| Command                    | Description                                                                                                                                                                                                   |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>write_rule_db</code> | Saves design netlist, constraints (including any incremental constraints) and litmus ruledb . This is the recommended command to save a Litmus session. The command <code>write_db</code> is not recommended. |
| <code>read_rule_db</code>  | Restores netlist, constraints and litmus ruledb. This is the recommended command to restore a Litmus session. The command <code>read_db</code> is not recommended.                                            |

 The commands are currently supported only in MMMC mode. It is not recommended to use them in Non-MMMC (`read_sdc`) mode as they have not been fully tested in that mode.

---

# Incremental Constraints with Restore Session

---

Litmus supports incremental constraints and rule check in a restored session from a saved DB.

Example Script:

```
# Load DB saved with write_rule_db from a earlier Litmus session
read_rule_db mysession.db

set_interactive_constraint_modes CM1

# Specify incremental constraint
set_clock_latency 0.3 [get_clocks CLK1]

# Perform incremental rule checking
verify_timing_intent

# Report incremental rule checks
report_rule_check -verbose
```

 The flow is currently supported only in MMMC mode. It is not recommended to use it in Non-MMMC (`read_sdc`) mode.

---

# Lint Configuration Support

---

Litmus can read lint configuration using the command `read_lint_configuration`. However, you need to manually create (or edit existing lint configuration file) as `write_lint_configuration` is currently not supported.

As Litmus is CUI only, therefore legacy mode commands from Conformal are not supported in Litmus. Following is a list (not comprehensive) of commands that are not supported by Conformal Litmus x and hence need to be excluded from lint configuration:

- `add_clock`
- `syntax`
- `unknown`
- `skipped`
- `set_wire_load_selection_group`
- `set_dont_touch`
- `set_dont_use`
- `set_guard_band_derate`
- `set_ideal_net`

Following commands have options that are different in Litmus than in Conformal Litmus. You need to update lint configuration accordingly: `set_wire_load_selection_group`

---

## Reduced Pessimism in Litmus

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- [CONST\\_CLK\\_GRP3](#)
- [CONST\\_DGN\\_CMB1](#) and [CONST\\_EXC\\_SMD7](#)
- [CONST\\_DGN\\_CMB3](#)
- [CONST\\_EXC\\_MCP5](#)
- [CONST\\_EXC\\_MCP7](#)

### CONST\_CLK\_GRP3

**Rule Description :** The harmonic cycle of a pair of connected and synchronous clocks exceeds the threshold

Reduced false violation in Litmus by skipping clock pairs which are synchronous but not connected, that is no path exists between the 2 clocks in the pair.



## Example

```
litmus > get_clock_to_clock_association clkA clkB
synchronous

litmus > report_timing -from clkA -to clkB -unconstrained
#####
#   Generated by:      Cadence Litmus
#   OS:               Linux x86_64 (Host ID xxxx)
#   Generated on:      xxx
#   Design:           test
#   Command:          report_timing -from clkA -to clkB -unconstrained
#####
No unconstrained timing paths with given description found.
Paths may be constrained or may not exist.

litmus > report_timing -from clkB -to clkA -unconstrained
#####
#   Generated by:      Cadence Litmus
#   OS:               Linux x86_64 (Host ID xxxx)
#   Generated on:      xxx
#   Design:           test
#   Command:          report_timing -from clkB -to clkA -unconstrained
#####
No unconstrained timing paths with given description found.
Paths may be constrained or may not exist.
```

Since no path exists for above clock pairs, no need to check for their harmonic threshold. No CONST\_CLK\_GRP3 violation is reported by Litmus

## CONST\_DGN\_CMB1 and CONST\_EXC\_SMD7

Litmus has split the existing rule CONST\_DGN\_CMB1 into two - CONST\_DGN\_CMB1 and CONST\_EXC\_SMD7

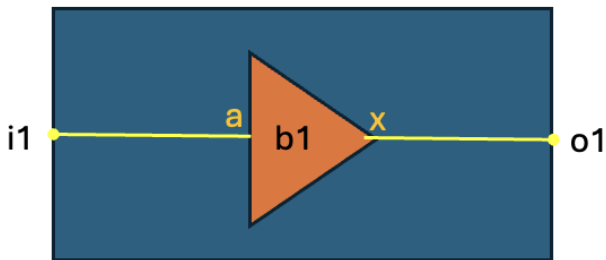
Rule Description of the new CONST\_DGN\_CMB1 now states the following: Unconstrained combinational path / Incomplete set\_min/max\_delay

If there is a combinational which cannot be constrained by use of one or more of the following constraints, CONST\_DGN\_CMB1 violation is not reported by Litmus:

- set\_input\_delay

- `set_output_delay`
- `set_max_delay`
- `set_min_delay`

## Example



```
create_clock -name clk -period 10 i1
set_propagated_clock clk
set_clock_sense -stop_propagation -clock clk b1/x
```

In the above design and constraints, there is no way to constraint the combinational path from port `i1` to `o1`. Hence, Litmus does not report a violation (Conformal Litmus does report violation). Other cases include the situation where the path is marked as false by `set_false_path`. So even if the user would constraint the path with `set_input_delay` or `set_max_delay`, it would be overridden by `set_false_path`. Hence Litmus does not report a violation.

## CONST\_DGN\_CMB3

### Rule Description : Inconsistent input/output delay value vs. clock period for combinational path

Case:

All combinational paths connecting the input/output port pair are marked false by `set_false_path`. Litmus ignores the `set_false_path` and reports a violation. However Litmus does not report a violation, since the inconsistency in IO delays is irrelevant.

## CONST\_EXC\_MCP5

### Rule Description: Incomplete multi-cycle path options

Reduced false violation by making the check based on timing intent, instead of constraint-based check and also skipped the check for globally ignored exceptions

### Example

#### top3.sdc

```
set_multicycle_path 4 -start -setup -from [get_clocks clk_cpu] -through [get_pins INST1/A] -to [get_clocks clk_cpu]
set_multicycle_path 3 -hold -from [get_clocks clk_cpu] -through [get_pins INST/A] -to [get_clocks clk_cpu]
```

Conformal Litmus reports two violations, one for each of the above two constraints, since each constraint is incomplete individually.

```
CONST_EXC_MCP5: Incomplete multi-cycle path options
Severity: Warning    Fail Occurrence: 1
1: Fail: In line 9, file top3.sdc (set_multicycle_path) : MCP definition
incomplete. Missing corners ' hold_rise hold_fall'
2: Fail: In line 10, file top3.sdc (set_multicycle_path) : MCP definition
incomplete. Missing corners ' setup_rise setup_fall'
```

Litmus does not report any violation since the overall timing intent is complete with the above two constraints applied.

## CONST\_EXC\_MCP7

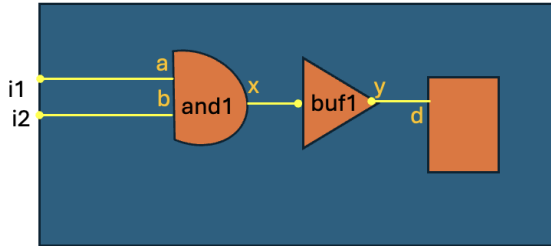
### Rule Description: Multi-cycle path start/end is on the lower frequency clock

Case: Globally ignored MCP exceptions

In case where the MCP exceptions are ignored due to existence exceptions of higher priority, this

violation is irrelevant. Hence it is skipped by Litmus (Conformal Litmus reports it).

## Example



```
set_multicycle_path 4 -setup -through and1/x
set_multicycle_path 3 -setup -through buf1/y
```

Violation for line number 1 above from Conformal Litmus is not reported in Litmus as the constraints at line number 2 overrides it.

## Rule Behavior Changes

- [CONST\\_CG\\_CHK3](#)
- [CONST\\_CLK\\_DEF1](#)
- [CONST\\_CLK\\_DEF2](#)
- [CONST\\_CLK\\_DEF40](#)
- [CONST\\_CLK\\_GRP2](#)
- [CONST\\_CLK\\_GRP4](#)
- [CONST\\_CLK\\_UNC10](#)
- [CONST\\_IO\\_DRC1](#)
- [CONST\\_IO\\_IDL1](#)
- [CONST\\_IO\\_IDL13](#)
- [CONST\\_IO\\_ITR1](#)

The following table explores the changes in various rule behavior from Conformal Litmus to Litmus.

| Rule          | Conformal Litmus                                                   | Litmus                                                                                                    | Additional Notes |
|---------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------------------|
| CONST_CG_CHK3 | Additionally considers generated clock / master clock association. | Checking is based on Tempus <code>check_timing</code> .<br><br>It thus checks the endpoint in the fanout. |                  |

**Litmus Migration Guide**  
Rule Behavior Changes--CONST\_EXC\_MCP5

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|                 |                                                                                                     |                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CONST_CLK_DEF1  | Covers the following scenarios:<br>MISSING_CLOCK,<br>BLOCKED_CLOCK,<br>MISSING_GENERATED_CLOCK      | Covers the following scenarios:<br>MISSING_CLOCK_NO_SHARED_DRIVER,<br>MISSING_CLOCK, BLOCKED_CLOCK<br><br>A new rule is introduced, namely<br>CONST_CLK_DEF4 which covers the<br>following case:<br>MISSING_GENERATED_CLOCK | Following behavior<br>difference between<br>Timing engines will<br>cause difference in<br>occurrence count:<br><br>Case when<br>create_generated_clock<br>-master_clock -> -<br>source path is missing :<br>Conformal Litmus:<br>Constraint object is not<br>created for rule check.<br>Litmus: Clock object is<br>created for rule check.<br>The clock object is not<br>included in clock<br>propagation and stops<br>others from being<br>propagated over the<br>reference.<br><br>Conformal Litmus: clock<br>is not propagated at<br>set_case_analysis<br>transition point.<br>Litmus: clock is<br>propagated at<br>set_case_analysis<br>transition point. |
| CONST_CLK_DEF2  | Extra violations found                                                                              | Fixed behavior                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| CONST_CLK_DEF40 | Constraint object is not created for<br>rule check.                                                 | Clock object is created for rule check.<br><br>The clock object is not included in clock<br>propagation and stops others from being<br>propagated over the reference.                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| CONST_CLK_GRP2  | Rule check is performed on<br>generated clocks in the transitive<br>fanout.                         | Rule Check is performed on immediate<br>generated clocks (directly inferred)                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| CONST_CLK_GRP4  | Rule check is based on root clock<br>source and clock group. Checked<br>on synchronous clock group. | Rule check is based on root clock and clock<br>relationship. Checked on synchronous but<br>not physically exclusive clock group.                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| CONST_CLK_UNC10 | Check performed on clock pairs<br>regardless of connection.                                         | Check performed on "connected" clock pairs<br>to see if they are asynchronous (w/o -<br>allow_path) or exclusive.                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                |                                                                                                                                                                                                                                                                                                                |                                                                                                                    |  |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--|
| CONST_IO_IDL1  | Only input ports that are connected to some constrained path end point (a clocked register or a pin with set_output_delay) are considered for this rule. However, a clock source will not be checked in this rule if it only drives clock pins of sequential elements.                                         |                                                                                                                    |  |
| CONST_IO_IDL13 | This rule applies only to ports that are start points of a timing path that does not match any set_false_path. set_false_path should have both -setup and -hold specified, or neither. Otherwise, the port is still subject to this rule check.<br><br>This rule does not check inputs that are clock sources. |                                                                                                                    |  |
| CONST_IO_ITR1  |                                                                                                                                                                                                                                                                                                                | Litmus skips clock source, which is checked by CTR1/CTR6<br><br>Litmus drops set_dont_touch_network for this check |  |

## CONST\_CG\_CHK3

- Litmus Additionally considers generated clock / master clock association.
- Litmus checking is based on Tempus check\_timing. It thus checks the endpoint in the fanout

## CONST\_CLK\_DEF1

In Litmus1 , CONST\_CLK\_DEF1 rule check covers the following scenarios:

1. MISSING\_CLOCK
2. BLOCKED\_CLOCK
3. MISSING\_GENERATED\_CLOCK

In Litmus, CONST\_CLK\_DEF1 rule check covers the following scenarios

1. MISSING\_CLOCK\_NO\_SHARED\_DRIVER (\*NEW)
2. MISSING\_CLOCK (other than #1)
3. BLOCKED\_CLOCK

A new rule is introduced in Litmus, namely CONST\_CLK\_DEF4 which covers the following case:

1. MISSING\_GENERATED\_CLOCK

Following behavior difference between Timing engines of Litmus & Litmus will cause difference in occurrence count:

Case when create\_generated\_clock -master\_clock -> -source path is missing :

Litmus: Constraint object is not created for rule check.

Litmus : Clock object is created for rule check. The clock object is not included in clock propagation and stops others from being propagated over the reference.

Litmus: clock is not propagated at set\_case\_analysis transition point.

Litmus: clock is propagated at set\_case\_analysis transition point.

## CONST\_CLK\_DEF2

### Rule Description : Unused clock constraint

**Issue:** Extra violations in Litmus as compared to Litmus

**Cause:** Bug in Litmus. Fixed in 23.20\_S500 and beyond.

#### Manifestation:

Case1 : A non-clock input pin of a library cell has a sequential arc to an output pin and the clock reaching the non-clock pin is launching the data to output. Litmus reports a CONST\_CLK\_DEF2 violation. Litmus does not report any violation which is the correct thing to do

## CONST\_CLK\_DEF40

Following behavior difference between Timing engines of Litmus & Litmus will cause difference in occurrence count:

Case when create\_generated\_clock -master\_clock -> -source path is missing :

Litmus: Constraint object is not created for rule check.

Litmus : Clock object is created for rule check. The clock object is not included in clock propagation and stops others from being propagated over the reference.



## CONST\_CLK\_GRP2

Litmus: Rule check is performed on generated clocks in the transitive fanout.

Litmus: Rule Check is performed on immediate generated clocks (directly inferred)

## CONST\_CLK\_GRP4

Litmus : Rule check is based on root clock source and clock group. Checked on synchronous clock group.

Litmus : Rule check is based on root clock and clock relationship. Checked on synchronous but not physically exclusive clock group.

## CONST\_CLK\_UNC10

Litmus : Check performed on clock pairs regardless of connection.

Litmus : Check performed on “connected” clock pairs to see if they are asynchronous (w/o -allow\_path) or exclusive.

## CONST\_IO\_DRC1

- Litmus(old version): Skips clock ports for this check
- Litmus(new version): Considers both clock ports as well as data ports for this check.

## CONST\_IO\_IDL1

- Only input ports that are connected to some constrained path end point (a clocked register or a pin with set\_output\_delay) are considered for this rule.  
However, a clock source will not be checked in this rule if it only drives clock pins of sequential elements.

## CONST\_IO\_IDL13

- This rule applies only to ports that are start points of a timing path that does not match any set\_false\_path. set\_false\_path should have both -setup and -hold specified, or neither. Otherwise, the port is still subject to this rule check.
- This rule does not check inputs that are clock sources.

## CONST\_IO\_ITR1

- Litmus skips clock source, which is checked by CTR1/CTR6
- Litmus drops set\_dont\_touch\_network for this check

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## New Rules in Litmus

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The following new rules are added in Litmus:

| Rule           | Category           | Description                                                                                                                                                         |
|----------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CONST_CLK_CTR2 | CONSTRAINT_SIGNOFF | set_clock_transition is not recommended for propagated clock                                                                                                        |
| CONST_CLK_DEF4 | CONSTRAINT_SIGNOFF | CONST_CLK_DEF1 from Litmus is split into CONST_CLK_DEF1 and CONST_CLK_DEF4.<br><br>CONST_CLK_DEF4 performs check for missing generated clocks where it is expected. |
| CONST_EXC_SMD7 | CONSTRAINT_SIGNOFF | Incomplete set_min_delay/set_max_delay exception for sequential path                                                                                                |

### 1D2S (1 Design, 2 SDC)

Litmus introduces a new feature called 1D2S (1 Design, 2 SDC). It enables you to compare two SDCs applied to the same design.

Refer to [Timing Constraint Comparison \(1D2S\)](#) in *Litmus Rule Reference* for all available rules.

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## Deprecated rules in Litmus

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Following rules available in Conformal Litmus are deprecated in Litmus:

| Rule            | Alternative Litmus Rule      | Reason for Deprecation |
|-----------------|------------------------------|------------------------|
| CONST_CG_CHK4   |                              |                        |
| CONST_DFT_CHK5  |                              |                        |
| CONST_DFT_CHK6  |                              |                        |
| CONST_DFT_CHK7  |                              |                        |
| CONST_DFT_CHK8  |                              |                        |
| CONST_CLK_CTR11 | Covered with CONST_CLK_CTR12 | Improved efficiency    |
| CONST_IO_ITR5   |                              |                        |
| CONST_IO_ITR6   |                              |                        |
| CONST_POW_CHK1  |                              |                        |
| CONST_POW_CHK2  |                              |                        |

# Launching GUI

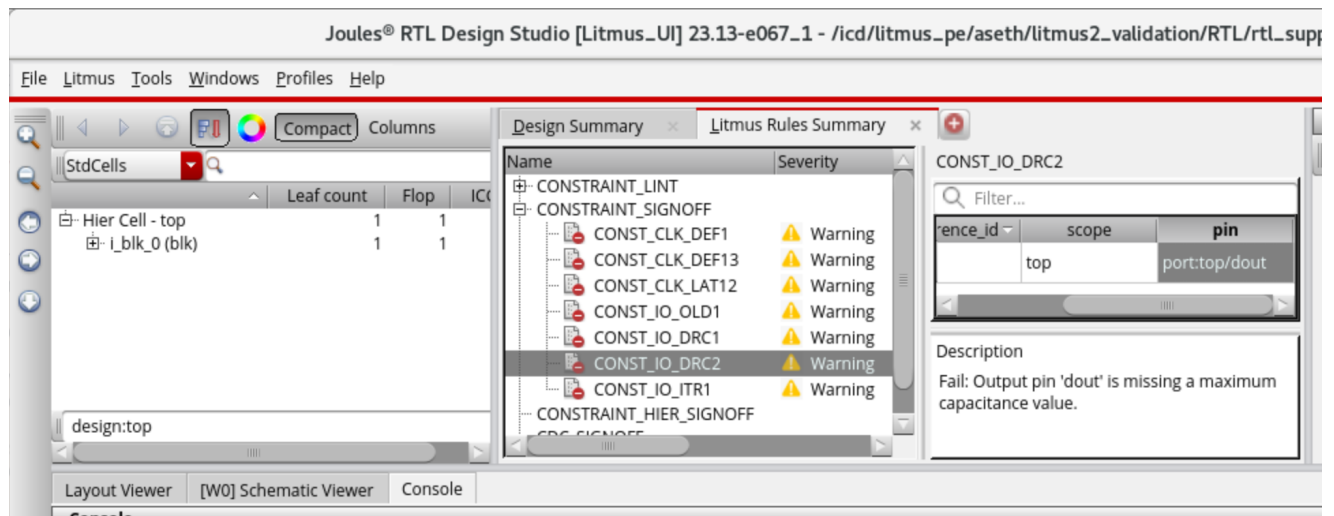
Litmus GUI is built on Cadence Joules Studio™ platform. In the current release, the GUI needs to be launched through Joules Studio shell.

## Steps to Launch GUI

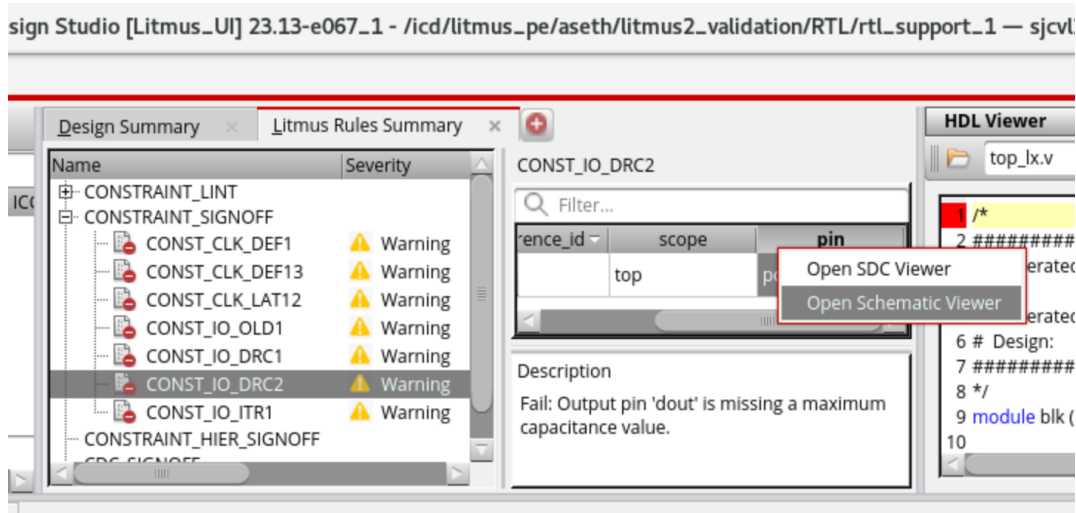
1. Ensure you have a Joules Studio installation
2. Before invoking Litmus, do
3. `setenv JOULES_STUDIO <joules_installation_path>`
4. In Litmus , after you have run `verify_timing_intent / check_target_constraints`, do the following:

`gui_show`

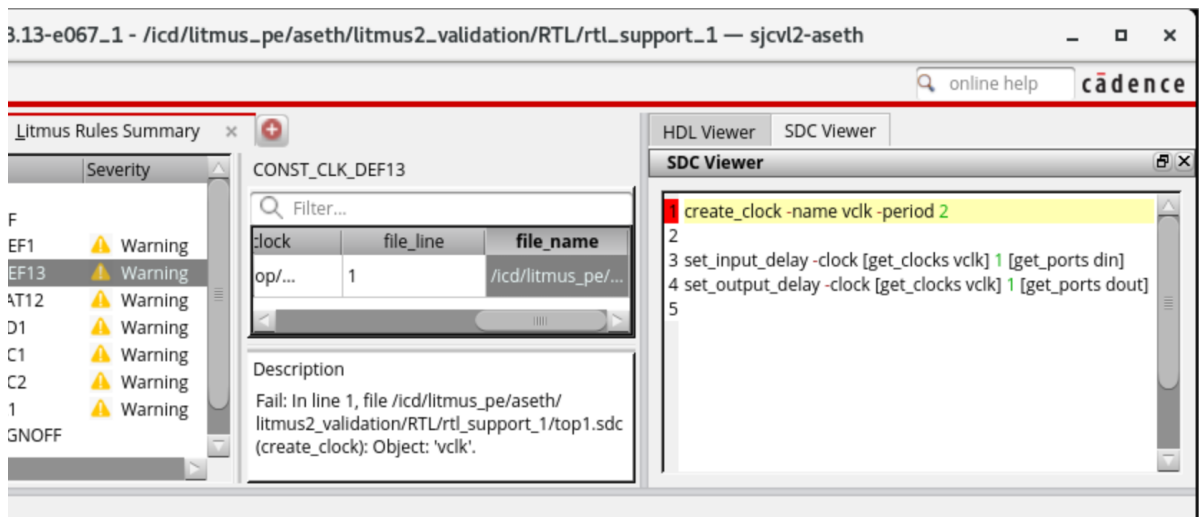
5. The Joules GUI should now show up in Litmus mode
6. Please note that `gui_show` should be done only after `init_design`
7. Rules check results should be visible in the "Litmus Rules Summary" tab.



8. Right click on rule occurrence row for Cross probing options.



- a. Click on *Open SDC viewer* to Cross probe to SDC file.

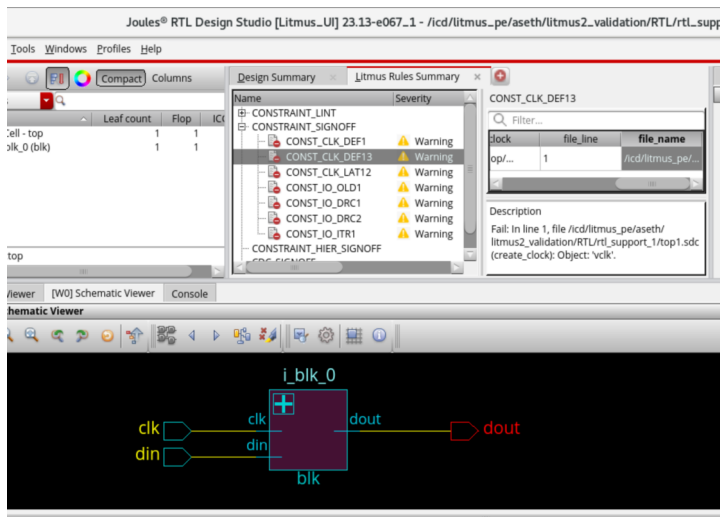


- b. Click on *Open Schematic viewer* to get the design schematic view.

## Litmus Migration Guide

### Launching GUI--Steps to Launch GUI

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**Note:** For RTL designs, only the elaborated netlist would be loaded in GUI and hence crossprobing back to RTL will not be available

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## Shell Default Behavior Changes

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Litmus shell has the following default behavior changes as compared to Litmus

| S No | Description                                                                                | Litmus setting to use to get behavior same as Litmus | Impact on Rule Checking |
|------|--------------------------------------------------------------------------------------------|------------------------------------------------------|-------------------------|
|      | report_timing reports unconstrained paths even when -unconstrained option is not specified | set_db<br>timing_report_unconstrained_paths<br>false | TODO                    |
|      |                                                                                            |                                                      |                         |
|      |                                                                                            |                                                      |                         |
|      |                                                                                            |                                                      |                         |
|      |                                                                                            |                                                      |                         |
|      |                                                                                            |                                                      |                         |
|      |                                                                                            |                                                      |                         |

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## SVA Generation

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### What is supported?

1. set\_multi\_cycle\_path
2. set\_false\_path

### What is not supported?

1. create\_generated\_clock
2. set\_case\_analysis
3. set\_disable\_timing
4. set\_clock\_sense

### Command Line Changes

1. There is no change in options for command "generate\_sva" in the current release
2. In Nov 15 release , following options will be available for new SVA system:
  - a. -use\_category : Enables new SVA system. When enabled , tool uses the -category\_file or SDC comment to write custom macro for SFP assertions
  - b. -category\_file : File containing categorization of SFP constraints



## How to enable Mode1 & Mode2

- This root attribute `ltms_sva_path_based_sampling_opt_level` controls the modes
- Only path based sampling is available in Litmus
- In the Nov 3 release Mode1 is enabled by default, no need for any setting. Traditional mode is not available in Nov 3 release
- For Mode2 do:
  - `set_db set ltms_sva_path_based_sampling_opt_level 2`

## Changes to control commands

|  | Litmus1 Command                                                                                                                                                                                                                                                                                                                    | Litmus equivalent                                                                                                                                                                                                                                                                                                    | Notes                                                                                                              |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|  | <code>set_sva_sfp_cycle &lt;num&gt;</code>                                                                                                                                                                                                                                                                                         | <code>set_db set_sva_sfp_cycle &lt;num&gt;</code>                                                                                                                                                                                                                                                                    |                                                                                                                    |
|  | <code>set_sva_to_generate</code><br><code>-file &lt;file_name&gt;</code><br><code>[&lt;number&gt;   -start_line &lt;number&gt; -end_line &lt;number&gt;]</code><br><br><code>set_sva_to_skip</code><br><code>-file &lt;file_name&gt;</code><br><code>[&lt;number&gt;   -start_line &lt;number&gt; -end_line &lt;number&gt;]</code> | <code>set_sva_to_generate</code><br><code>-file &lt;file_name&gt;[:number]  </code><br><code>-start_line &lt;number&gt; -end_line &lt;number&gt;]</code><br><br><code>set_sva_to_skip</code><br><code>-file &lt;file_name&gt;[:number]  </code><br><code>-start_line &lt;number&gt; -end_line &lt;number&gt;]</code> | Litmus:<br><code>-file mysdc:43</code><br>In Litmus1 it is <code>-file mysdc 43</code><br>(without the colon)      |
|  | <code>set_sva_generate_comment</code><br><br><code>&lt;comment&gt;</code>                                                                                                                                                                                                                                                          | <code>set_db</code><br><code>ltms_sva_generate_comment</code><br><code>&lt;comment&gt;</code>                                                                                                                                                                                                                        | Assertions will be generated only for those SFP/MCP whose SDC comment matches the comment specified with this attr |

|                                                                                                      |                                                                                                                                                                                                      |                                                                                                                                     |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <p>set_sva_assertion_prefix<br/>[-sfp &lt;prefix&gt;   -mcp<br/>&lt;prefix&gt;   &lt;prefix&gt;]</p> | <p>set_db<br/>ltms_sva_sfp_assertion_prefix<br/>&lt;prefix&gt;</p> <p>set_db<br/>ltms_sva_mcp_assertion_prefix<br/>&lt;prefix&gt;</p> <p>set_db<br/>ltms_sva_assertion_prefix<br/>&lt;prefix&gt;</p> | <p>ltms_sva_sfp_assertion_prefix<br/>and<br/>ltms_sva_mcp_assertion_prefix<br/>have priority over<br/>ltms_sva_assertion_prefix</p> |
| <p>set_sva_assertion_suffix<br/>[-sfp &lt;suffix&gt;   -mcp<br/>&lt;suffix&gt;   &lt;suffix&gt;]</p> | <p>set_db<br/>ltms_sva_sfp_assertion_suffix<br/>&lt;suffix&gt;</p> <p>set_db<br/>ltms_sva_mcp_assertion_suffix<br/>&lt;suffix&gt;</p> <p>set_db<br/>ltms_sva_assertion_prefix<br/>&lt;prefix&gt;</p> | <p>ltms_sva_sfp_assertion_suffix<br/>and<br/>ltms_sva_mcp_assertion_suffix<br/>have priority over<br/>ltms_sva_assertion_suffix</p> |
| <p>set_sva_module_prefix<br/>[-sfp &lt;prefix&gt;   -mcp<br/>&lt;prefix&gt;   &lt;prefix&gt;]</p>    | <p>set_db<br/>ltms_sva_sfp_module_prefix<br/>&lt;prefix&gt;</p> <p>set_db<br/>ltms_sva_mcp_module_prefix<br/>&lt;prefix&gt;</p> <p>set_db<br/>ltms_sva_module_prefix<br/>&lt;prefix&gt;</p>          |                                                                                                                                     |
| <p>set_sva_module_suffix<br/>[-sfp &lt;suffix&gt;   -mcp<br/>&lt;suffix&gt;   &lt;suffix&gt;]</p>    | <p>set_db<br/>ltms_sva_sfp_module_suffix<br/>&lt;suffix&gt;</p> <p>set_db<br/>ltms_sva_mcp_module_suffix<br/>&lt;suffix&gt;</p> <p>set_db ltms_sva_module_suffix<br/>&lt;suffix&gt;</p>              |                                                                                                                                     |

|  |                                                                                                  |                                                                                                                                                  |                             |
|--|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
|  | set_sva_file_prefix<br>[-sfp <prefix>   -mcp<br><prefix>   -cgc <prefix>  <br><prefix>]          | set_db ltms_sva_file_prefix<br><prefix><br><br>set_db ltms_sva_sfp_file_prefix<br><prefix><br><br>set_db<br>ltms_sva_mcp_file_prefix<br><prefix> |                             |
|  | set_sva_file_suffix<br>[-sfp <suffix>   -mcp <suffix><br>  -cgc <suffix>   <suffix>]             | set_db ltms_sva_file_suffix<br><suffix><br><br>set_db ltms_mcp_file_suffix<br><suffix><br><br>set_db ltms_file_suffix <suffix>                   |                             |
|  | set_sva_reset_signal_name<br>[-sfp <name>   -mcp <name><br>  -cgc <name>   -unique  <br><name> ] | set_db<br>ltms_sva_reset_signal_name<br><signal_name>                                                                                            | At -unique is not supported |